```
config system npu
    config fp-anomaly
        set sctp-csum-err {allow | drop | trap-to-host}
    end
end
```

`allow`	The NP7 platform will forward all SCTP packets with zero checksum.
`drop`	The NP7 platform will drop all SCTP packets with zero checksum. This is the default value.
`trap-to-host`	NP7 anomaly protection for SCTP will be disabled and SCTP packets with zero checksum will be forwarded. This option can also be used with a DoS policy to configure anomaly protection. If `policy-offload-level` is set to `dos-offload`, DoS policy anomaly protection is offloaded to the NP7 platform.

Industrial connectivity

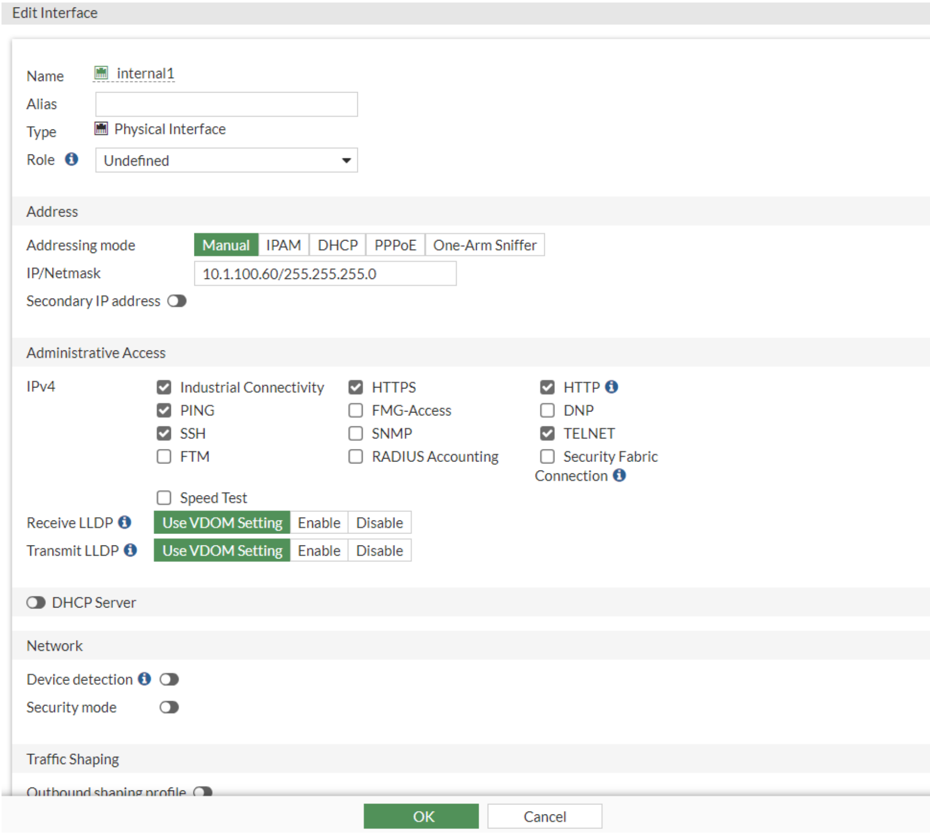
The industrial connectivity daemon (icond) and Industrial Connectivity service are available for FortiGate Rugged models equipped with a serial RS-232 (DB9/RJ45) interface to:

- Receive data in IEC 60870-5-101 serial protocol and convert it to IEC 60870-5-104 TCP/IP. See Sample configuration to convert IEC 60870-5-101 serial to IEC 60870-5-104 TCP/IP transport on page 1052.
- Receive data in Modbus serial (RTU/ASCII) protocol and convert it to Modbus TCP. See Sample configuration to convert Modbus serial to Modbus TCP on page 1053.

You can allow Industrial Connectivity access to an interface in the GUI and CLI.

To enable Industrial Connectivity for an interface in the GUI:

1. Go to *Network > Interfaces*.
2. Click *Create New > Interface*, or double-click an interface to open in for editing.
3. Set *Role* to *Undefined* or *WAN*
   Only internal and WAN interfaces support *Industrial Connectivity* administrative access.
4. In the *Administrative Access* section, select *Industrial Connectivity*.



5. Set the remaining options as desired, and click *OK*.

To enable Industrial Connectivity for an interface in the CLI:

```
config system interface
    edit <name>
        set allowaccess icond
        ...
    next
end
```

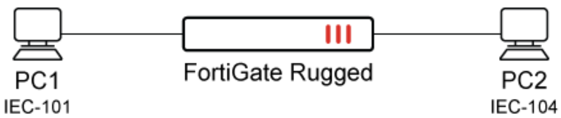
`allowaccess icond`	Specify what types of management protocols can access the interface: • `icond`: Industrial Connectivity service access to proxy traffic between serial port and TCP/IP.

Use the `config system icond` command to configure the Industrial Connectivity service provided by the Industrial Connectivity daemon (icond).

Sample configuration to convert IEC 60870-5-101 serial to IEC 60870-5-104 TCP/IP transport

After the Industrial Connectivity service is enabled and configured for an interface, supported FortiGate Rugged devices can receive data from Supervisory Control and Data Acquisition (SCADA) systems in serial format and convert it to TCP/IP formats.

In the following topology, PC1 uses the IEC 60870-5-101 (shortened to IEC-101) protocol to transmit data from SCADA systems to FortiGate Rugged, where the data is converted to the IEC 60870-5-104 (shortened to IEC-104) protocol, and sent to PC2.



The data is converted as follows:

- FortiGate Rugged transmits data over TCP port 502.
- Protocols IEC 60870-5-101 and IEC 60870-5-104 are both used to transmit the data.

  While IEC-101 is based on a serial transmission of data (for example, using RS-232 and FSK-based modems), IEC-104 is packet oriented and based on TCP/IP transmission.

To enable Industrial Connectivity for an interface in the CLI:

```
config system interface
    edit "internal1"
        set vdom "root"
        set ip 10.1.100.60 255.255.255.0
        set allowaccess ping https ssh http telnet icond
        set type physical
        set snmp-index 3
    next
end
```

To configure the Industrial Connectivity service in the CLI:

1. Configure the Industrial Connectivity service

```
config system icond
    set status enable
    set type iec101-104
    set tty-device "serial0"
end
```

2. Get the default status:

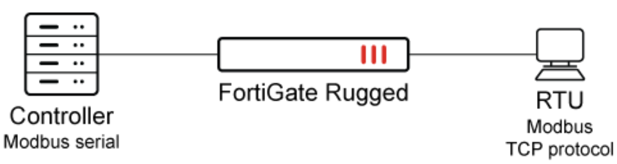
```
get system icond
status              : enable
type                : iec101-104
tty-device          : serial0
```

```
tty-baudrate        : 9600
tty-parity          : even
tty-databits        : 8
tty-stopbits        : 1
tty-flowcontrol     : none
iec101-mode         : unbalanced
iec101-laddr-size   : 1
iec101-laddr-local  : 1
iec101-laddr-remote : 2
iec101-use-ack-char : disable
iec101-keepalive    : enable
iec101-t0           : 500
iec101-trp          : 2500
iec104-t1           : 15
iec104-t2           : 10
iec104-t3           : 20
iec104-k            : 12
iec104-w            : 8
```

Sample configuration to convert Modbus serial to Modbus TCP

After the Industrial Connectivity service is enabled and configured for an interface, supported FortiGate Rugged devices can receive data from Modbus networks in serial format and convert it to TCP.

In the following topology, the Modbus controller uses Modbus serial to transmit data to FortiGate Rugged, where the data is converted to the Modbus TCP, and sent to the Modbus Remote Terminal Unit (RTU).



To enable Industrial Connectivity for an interface in the CLI:

```
config system interface
    edit "internal1"
        set vdom "root"
        set ip 10.1.100.60 255.255.255.0
        set allowaccess ping https http telnet icond
        set type physical
        set description "link to modbus server"
        set snmp-index 3
    next
end
```